Demonstration of Network Communication Models (OSI / TCP-IP) Using a Network Simulation Tool

Case Study 4

Aim

To demonstrate the OSI and TCP/IP network communication models by building and testing a small network in a simulation tool.

Objective

- Understand the layers of the OSI and TCP/IP models.
- Build a network topology using Cisco Packet Tracer.
- Observe how data is encapsulated and processed layer by layer.
- Verify connectivity using ping and Simulation mode.

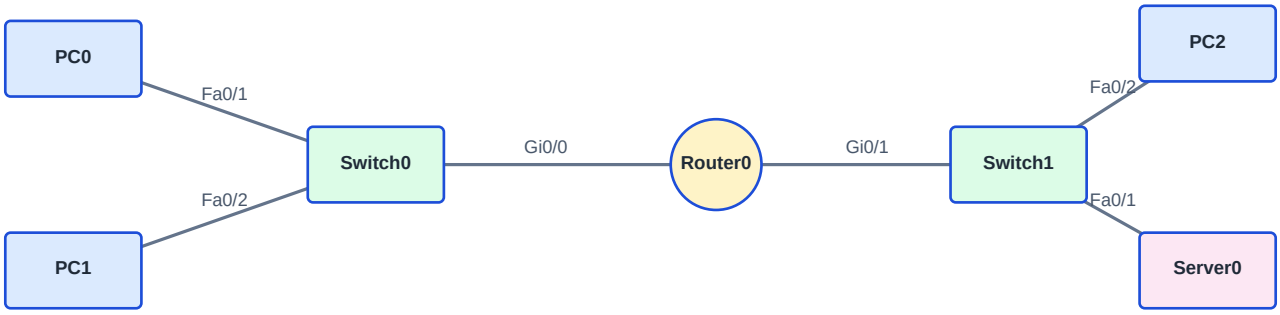
Tool Used

Cisco Packet Tracer (free for learners through Cisco Networking Academy). Equivalent alternatives: GNS3, Eve-NG.

OSI and TCP/IP Models

OSI Layer	TCP/IP Layer	PDU	Function	Examples
7 Application	Application	Data	User services	HTTP, DNS, FTP, SMTP
6 Presentation	Application	Data	Encoding, encryption, compression	SSL/TLS, JPEG
5 Session	Application	Data	Session setup and control	NetBIOS, RPC
4 Transport	Transport	Segment	End-to-end delivery, ports	TCP, UDP
3 Network	Internet	Packet	Logical addressing, routing	IP, ICMP, Router
2 Data Link	Network Access	Frame	Physical addressing (MAC), switching	Ethernet, Switch
1 Physical	Network Access	Bits	Signal transmission	Cables, hubs

Network Topology



IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.11	255.255.255.0	192.168.1.1
Router0	Gi0/0	192.168.1.1	255.255.255.0	–
Router0	Gi0/1	192.168.2.1	255.255.255.0	–
Server0	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC2	NIC	192.168.2.11	255.255.255.0	192.168.2.1

Procedure

1. Open Packet Tracer and place 3 PCs, 1 server, 2 switches and 1 router.
2. Connect devices with copper straight-through cables as per the topology.
3. Assign IP address, mask and gateway to every PC and the server (Desktop → IP Configuration).
4. Configure the router interfaces:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# no shutdown
```

1. Test connectivity from PC0 command prompt: `ping 192.168.2.10`.
2. Switch to **Simulation mode**, filter for ICMP / HTTP / DNS, and send a PDU from PC0 to Server0.
3. Click each device's event and open **OSI Model** tab to inspect how each layer handles the PDU.
4. Open PC0's web browser and access the server's IP address to see HTTP over TCP.

Observations

Observation	Layer(s) Involved
Ping succeeds after the router is configured (ARP then ICMP echo).	Network, Data Link
PC encapsulates data: Data → Segment → Packet → Frame → Bits.	Transport, Network, Data Link, Physical
Switch forwards frames using the MAC address table; it does not change the IP header.	Data Link
Router removes the frame, reads the destination IP, selects the route and builds a new frame.	Network
HTTP request uses TCP port 80 with a three-way handshake (SYN, SYN-ACK, ACK).	Application, Transport

Note: Add your own Packet Tracer screenshots (topology, ping result and Simulation-mode OSI tab) beside the observation table.

Results

- The network was built successfully and end-to-end communication between both subnets was verified.
- Encapsulation and de-encapsulation at each OSI layer were observed in Simulation mode.
- The mapping of the 7-layer OSI model to the 4-layer TCP/IP model was demonstrated.

Conclusion

Network simulation tools provide a safe way to understand layered communication. The OSI model is a reference for learning, while the TCP/IP model is the practical protocol suite used on the Internet. Simulating both clarified how data moves from application to physical media and back.